Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

Report on Extra-Curricular activities done as a part of

EXC-1033 Robotics for Engineers

Submitted by

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21BEC0996

To

Prof. Rashmi Ranjan Das

Winter Semester, 2023-2024

S. No	Event & Activities Name	Date	Hours Spent	Page No.
1	**360° with FUSION**	9th JUNE, 2023	2.5	3
2	**TINKERING WITH ARDUINO**	11th June 2023	2.5	4
3	**360° with FUSION Part 2**	15th June 2023	2.5	5
4	**PI EXPLORATION**	19TH JUNE 2023	2.5	6
5	**IC FUSION: POWER OF PCBS**	25TH JUNE 2023	2.5	7
6	**EXPLORING NLP IN ROBOTICS**	16TH July 2023	2	8
7	**Programming in C and Data Structures and Algorithms**	20th JUNE 2023	10	9
8	**Kinematics**	15TH July 2023	10	9
9	**Manufacturing Processes and Materials**	5th AUGUST 2023	10	9
10	**Final Assessment Test**	5th AUGUST 2023	3	9
11	**Project**	7th AUGUST 2023	48	10
12	**Report preparation**	9th AUGUST 2023	5	

		Total Hours	**110.5**	

Description of Events/Activities

1. 360° with FUSION

Date: 9th JUNE, 2023

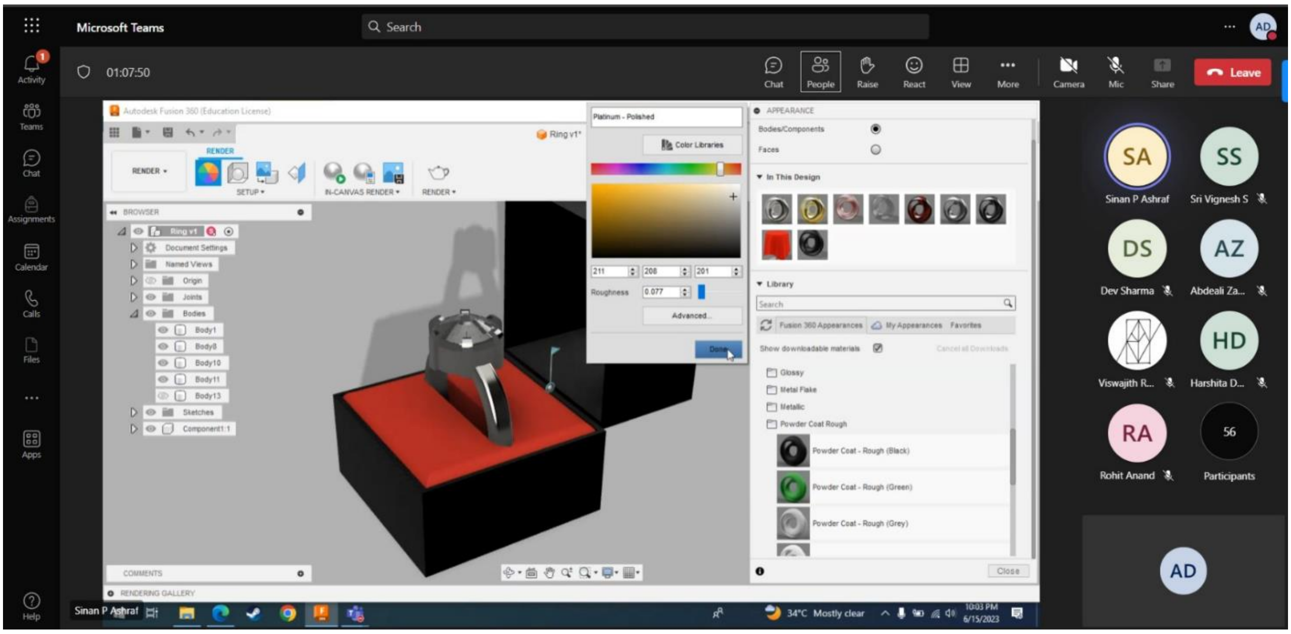
Number of Participants: 75

The Introduction to Fusion 360 event, organized by Swati and Sinan, aimed to familiarize participants with 2D sketching and 3D modelling using Fusion 360 software. The event was attended by a diverse group of individuals who were interested in learning about the software.

The event started with an introduction to Fusion 360 by Sinan, who provided a brief overview of the interface, which ranged from 2D sketching to 3D modelling, and its functionality in the product design and manufacturing industry. Swati then took over the presentation and went into detail about the features and functionalities of Fusion 360. Sinan explained that Fusion 360 is a comprehensive tool that covers a wide range of functionalities, from 3D modelling to simulation, collaboration, and manufacturing. He then proceeded to demonstrate how to create a 3D model of a product, component, and assembly using the software was demonstrated by Swati.

The importance of collaboration in product design and manufacturing was highlighted and showed how Fusion 360 allows for easy collaboration between team members. The event was very informative and interactive, with the speakers answering questions from the audience throughout the presentation.

In conclusion, the DE-FUSION event was a success, providing attendees with a first-hand experience of the industry-leading software. The event was an excellent opportunity for anyone interested in product design and manufacturing to learn about the basics of Fusion 360 and its various functionalities.



2. <u>Tinkering with Arduino</u>

Date: 11th June 2023

Number of Participants: 118

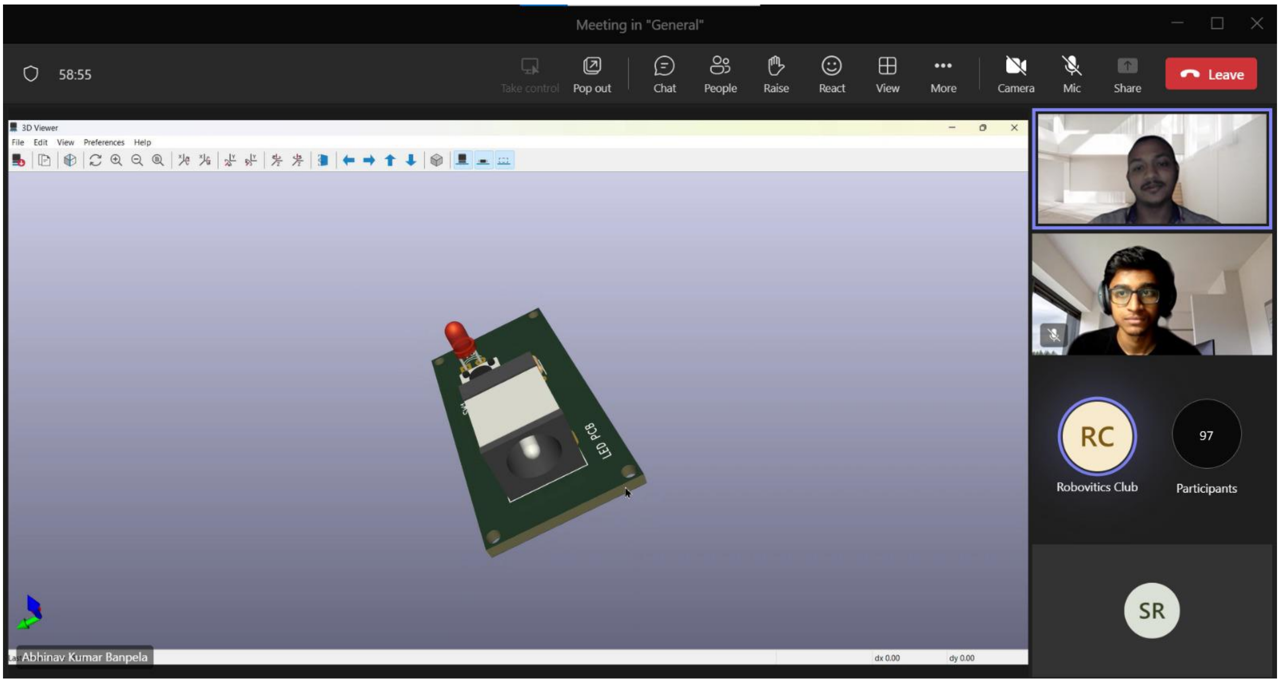
EVENT COMPLETION SUMMARY:

The Tinkering with Arduino speaker session provided participants with a comprehensive understanding of Arduino and its practical applications. The event covered various sensors and their real-world applications, explained the difference between microprocessors and microcontrollers, and introduced the components of Arduino Uno and the Arduino IDE. Simulations on Tinkercad allowed participants to learn how to program Arduino to perform tasks like blinking LEDs, controlling LED brightness using PWM, and interfacing with sensors such as PIR and ultrasonic sensors. Overall, the session equipped participants with the knowledge and skills to implement Arduino projects and explore the world of electronics and prototyping.

3. 360° with FUSION Part-2

Date: 15th June 2023

Number of Participants: 85

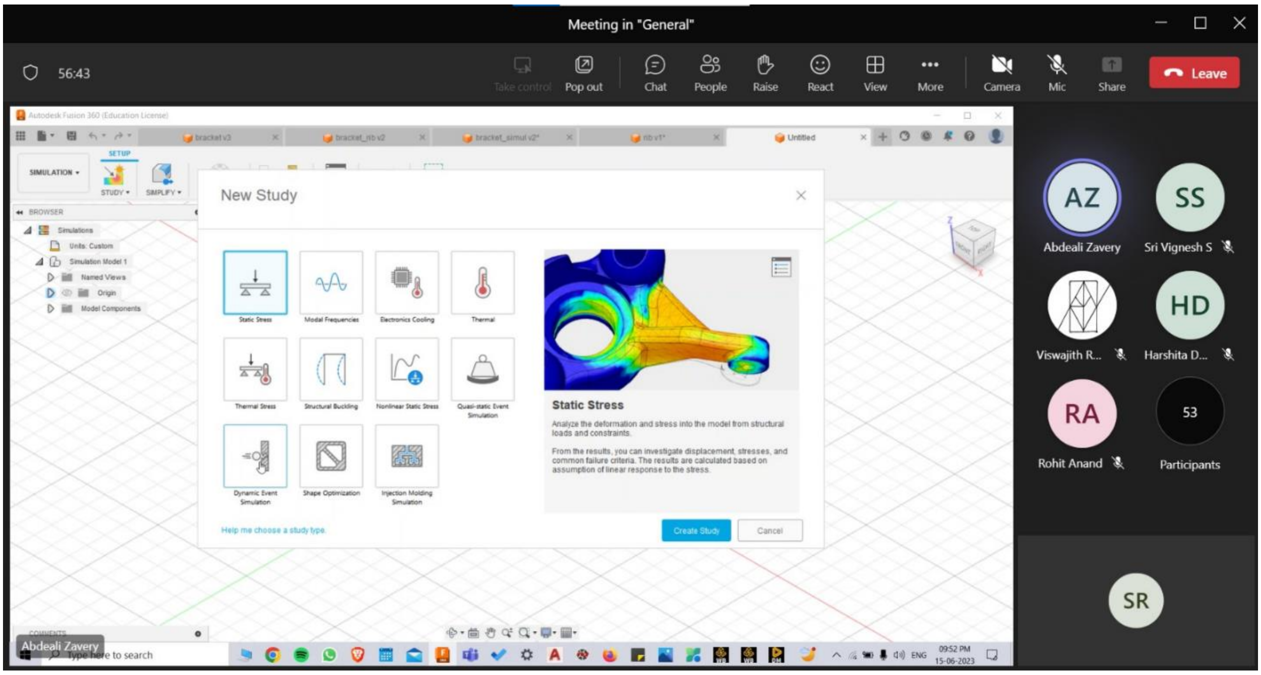


Event Completion Summary:

The EXC event organized by RoboVITics commenced on schedule at 9:00 PM on Thursday, 15th June 2023. This event aimed to provide participants with a comprehensive understanding of Fusion 360 software and its simulation and rendering capabilities. The first speaker, Abdeali Zavery, guided the participants through the process of setting up simulations, defining material properties, applying boundary conditions, and interpreting simulation results. The session was also very interactive with regular questions and doubts asked. The next speaker, Sinan P Ashraf, provided demonstrations on setting up scenes, adjusting lighting conditions, and applying materials to enhance the visual quality of the rendered image.

The event ended with a Q&A session where participants had the opportunity to ask Swati and Sinan any questions, they had about Fusion 360. The speakers provided valuable insights and suggestions.

Overall, 360 with Fusion Part-2 was an informative event that provided participants with a comprehensive understanding of Fusion 360 and practical tips on how to use it effectively.



4. PI EXPLORATION

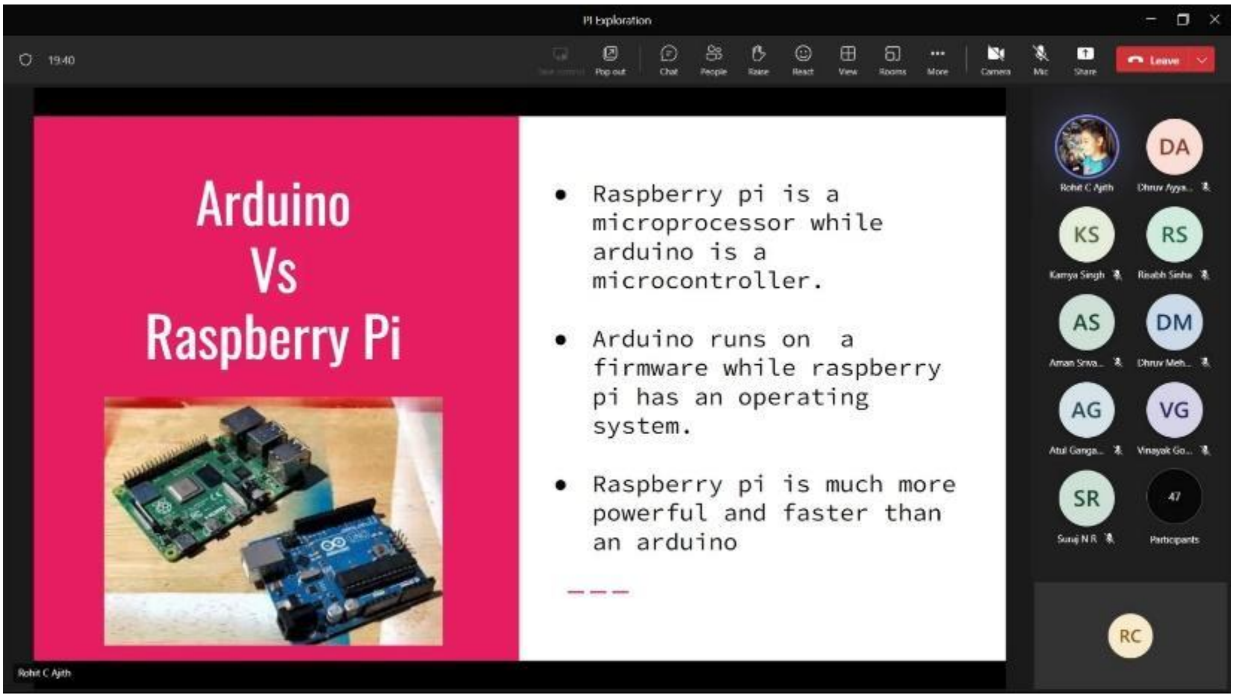
Date: 19TH JUNE 2023

Number of Participants: 104

The EXC event organized by RoboVITics commenced on schedule at 9:00 PM on Monday, 19th June 2023. It was a comprehensive event that delved into the intricacies of Raspberry Pi, providing participants with a detailed introduction to the device, exploring its wide range of applications across various domains, discussing different models available in the market, offering step-by-step guidance on setting up the Raspberry Pi, familiarizing attendees with terminal navigation and headless mode, teaching basic coding skills using Python, and concluding with inspiring examples of Raspberry Pi projects. This enriching workshop equipped participants with the necessary knowledge and expertise to unlock the full potential of Raspberry Pi, empowering them to pursue their own innovative projects with confidence and creativity.

To acquire a basic understanding of how to programme the Raspberry Pi using Python, two straightforward programmes were described: blinking an LED and motion detection using a PIR sensor and a piezoelectric buzzer. At the end, a video compilation of several fascinating and extremely practical Raspberry Pi projects from around the world was exhibited.

Over the course of the event, the speaker engaged with the audience and addressed any questions they may have had.



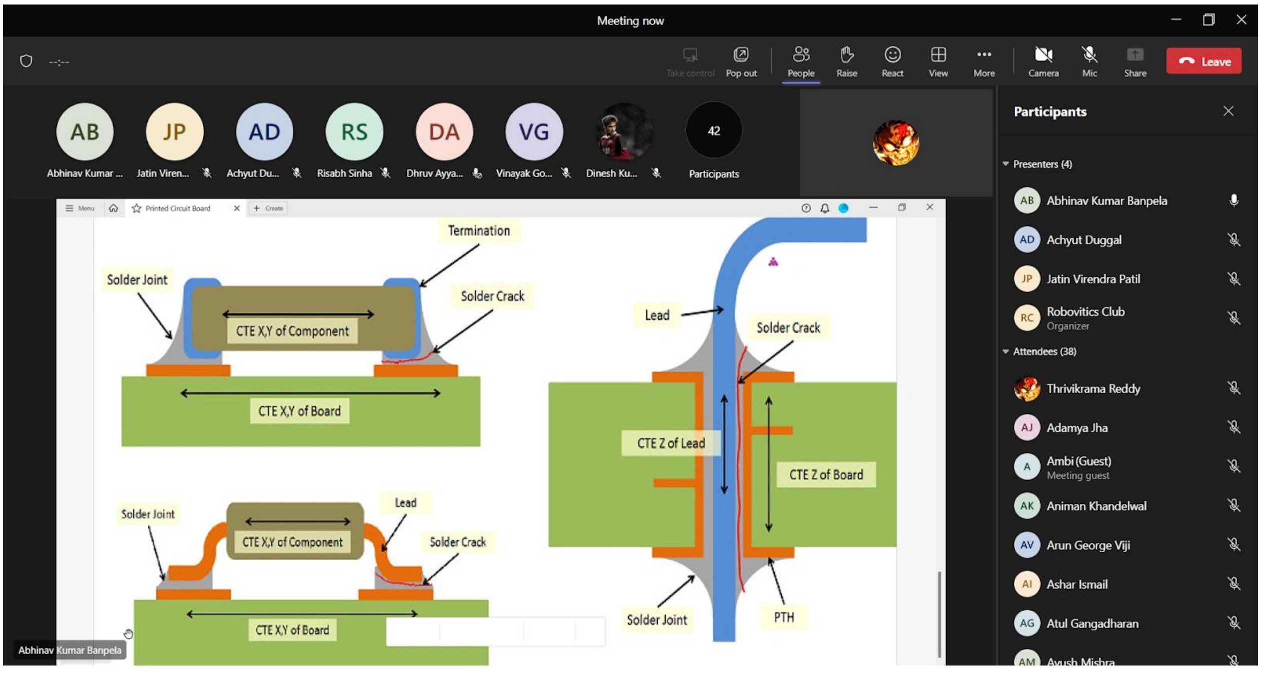
5. IC FUSION: POWER OF PCBS

Date: 25TH JUNE 2023

Number of Participants: 140

The EXC event organized by RoboVITics commenced on schedule at 6:00 PM on Sunday, 25th June 2023. This event focused on explaining about basics of PCBs. The basic difference between through-hole and SMD components used on PCBs was explained in detail. In this session, the KiCad Software is used. All the essential tools were explained briefly. Schematic for a basic Led Circuit was designed, later the process for converting the schematic to PCB was explained in detail. The basic workings of the PCB editor were explained in this process, how components should be placed on the board, how tracing should be done, and, how components should be connected. Finally, the creation of the Gerber file was explained and then all the doubts were clarified at the end of the session.

By the end of the session, participants had gained a comprehensive understanding of PCBs and ICs, as well as the skills and knowledge necessary to design and develop their own PCBs and ICs. Overall, the "ICing Electronics with PCB" event was a success and provided valuable insight into the design and development of electronic components.



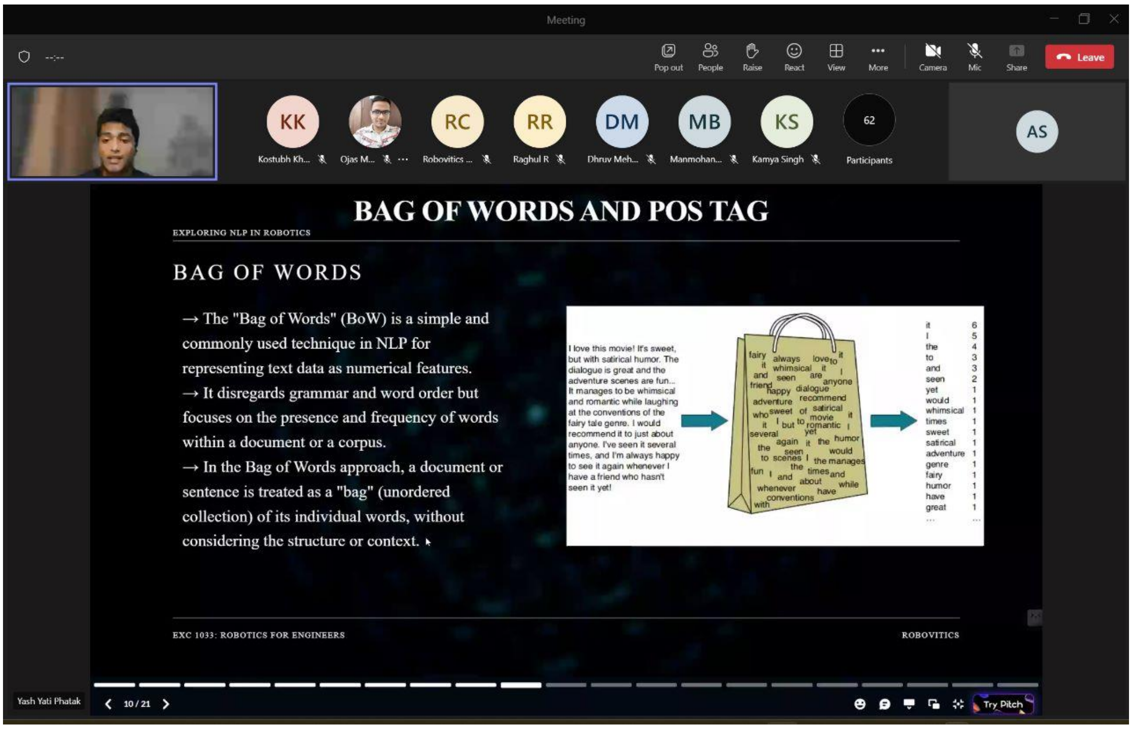
6. EXPLORING NLP IN ROBOTICS

Date: 16TH July 2023

Number of Participants: 110

The EXC event organized by RoboVITics on July 16, 2023, focused on the basics of NLP (Natural Language Processing). The speakers for our event were Mr. Yash Phatak and Mr. Ojas Mittal, who is a core committee member of the RoboVITics club.

The event covered topics such as tokenization, text cleaning, and lemmatization, demonstrating their processes with examples. The Bag of Words model was extensively taught, and the role of POST TAG was briefly explained. The event also highlighted the distinctions between stemming and lemmatization, providing effective strategies for their usage. Several commonly used NLP tools and libraries were showcased and thoroughly discussed. The applications of NLP and its latest technologies were explored. A speech recognition program employing various NLP libraries and tools was demonstrated, with its functionality explained. Attendees received the program's code for further practice, and the event concluded with a Q&A session.



7. Programming in C and DSA

Date: 20th JUNE 2023

Number of Participants: 80

RoboVITics distributed a study material named ‘C and DSA’ that covered various topics ranging from basic C to searching and sorting, and elaborated on linked lists, stacks, and queues. The primary aim of the material was to assist beginners in learning these topics. Participants were required to read the material for a week and then take a quiz that covered all aspects of the material, from basic to advanced concepts.

8. Kinematics

Date: 15TH July 2023

Number of Participants: 80

RoboVITics provided a study material called ' Kinematics' that covered topics such as Frames, Articulated Robots, translation and rotation of frames, combined transformations, inverse of transformation matrix, forward and inverse kinematics of robots, DH representation, and differential bots. The aim of the material was to assist beginners in learning these topics. Participants were expected to read the material for a week and take a quiz that tested their understanding of the material, covering both basic and advanced concepts.

9. Manufacturing Processes and Materials

Date: 5th AUGUST 2023

Number of Participants: 80

RoboVITics provided a study material called 'Manufacturing Processes and Materials' that covered topics such as Introduction to Manufacturing, Milling, Metal Casting, Materials for Pattern Making, and Drilling. The aim of the material was to assist beginners in learning these topics. Participants were expected to read the material for a week and take a quiz that tested their understanding of the material, covering both basic and advanced concepts.

10. Final Quiz

Date: 5th AUGUST 2023

Number of Participants: 80

RoboVITics distributed the hyperlink to the final quiz assignment for students to partake. The quiz encompassed questions spanning all topics taught throughout the course, and students were duly briefed on the pertinent rules and instructions. The students were accorded the liberty to attempt the quiz at their own discretion during the entire duration of the day.

11. **Project**

Date: 7^{th} AUGUST 2023

Title: Formulating an Adaptive Robot Gripper for Improved Object Manipulation'

The goal of this project is to use Fusion 360 to design and create an adaptive robot gripper that overcomes the constraints of traditional grippers in terms of variety and grasping ability. Automation operations are less efficient since the current robot grippers frequently struggle to grasp items of various sizes and forms. Robotics has a huge problem in developing a gripper that can adapt to various objects while keeping a firm hold.